AWS PRODUCTION ARCHITECTURE

AWS Hands-On Cloud Infrastructure Project

Project Type: AWS Cloud / Networking / Load Balancing / Auto Scaling

AWS Region: Asia Pacific (Mumbai) — ap-south-1

VPC: VPC-A

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Project Components :

- Amazon VPC
- Public and Private Subnets
- Internet Gateway
- NAT Gateways
- Route Tables
- Security Groups
- Amazon EC2
- EC2 Launch Template
- Application Load Balancer
- Target Group
- Auto Scaling Group

Project Purpose

This project documents a highly available AWS web application infrastructure designed using an Application Load Balancer, EC2 instances, Auto Scaling, public and private subnets, and NAT Gateways.

1. Project Overview

The **AWS Application Load Balancer with Auto Scaling** project demonstrates the design of a distributed web application infrastructure inside an Amazon VPC. The infrastructure is deployed in the **ap-south-1 Mumbai Region** and uses three Availability Zones:

- ap-south-1a
- ap-south-1b
- ap-south-1c

The VPC contains six subnets divided into three public and three private subnets. The public subnets are used for internet-facing components such as the Application Load Balancer, while the private subnets are designed to host EC2 application instances.

Three NAT Gateways provide outbound Internet connectivity for the private subnets.

The Application Load Balancer distributes incoming HTTP requests to EC2 instances registered through a Target Group.

2. Project Objective

The main objectives of the project are:

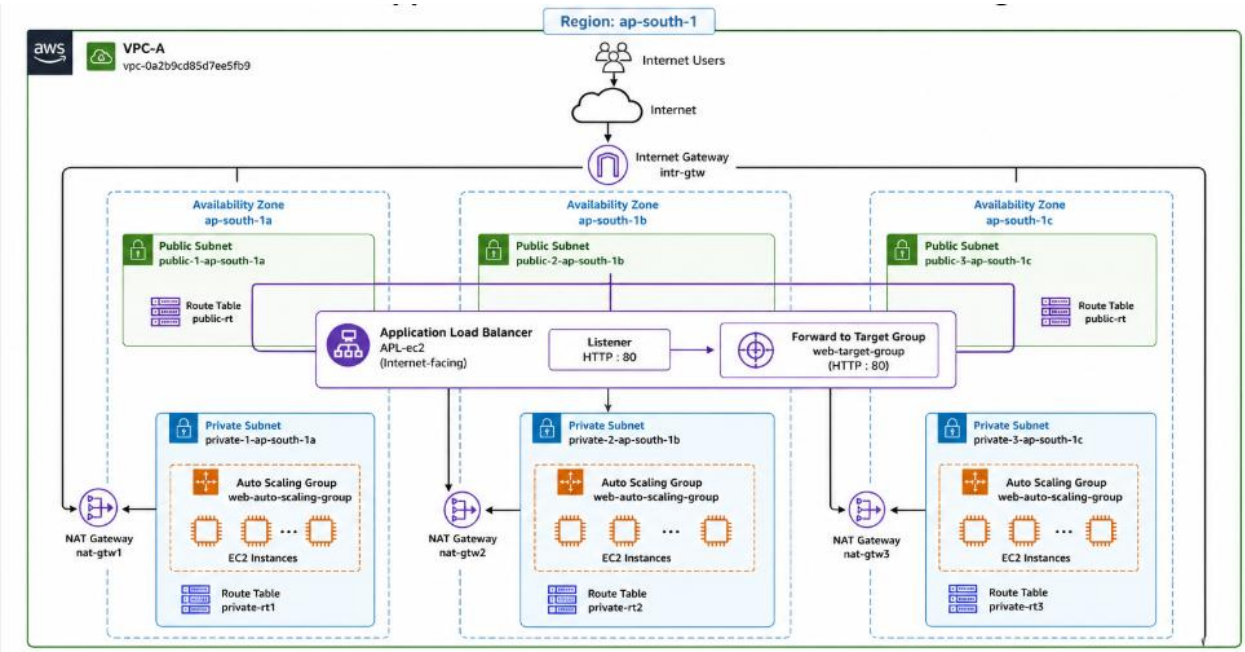
1. Design a multi-AZ AWS network.
2. Separate public and private resources.
3. Deploy an Internet-facing Application Load Balancer.
4. Use a Target Group for EC2 traffic distribution.
5. Use an EC2 Launch Template for standardized instance configuration.
6. Integrate EC2 instances with an Auto Scaling Group.
7. Provide outbound Internet access to private EC2 instances through NAT Gateways.
8. Implement health checks for application targets.
9. Understand high availability and load balancing.

3. Architecture

The project architecture consists of an AWS VPC spanning three Availability Zones.
Each Availability Zone contains:

- One public subnet
- One private subnet

The Application Load Balancer is deployed across the three public subnets.
The application EC2 instances are placed in private subnets and are associated with the Target Group and Auto Scaling Group.



4. VPC Configuration

Property	Actual configuration
VPC name	VPC-A
VPC ID	vpc-0a2b9cd85d7ee5fb9
Region	ap-south-1
IPv4 CIDR	10.0.0.0/16
State	Available
DNS resolution	Enabled
DNS hostnames	Disabled
Default VPC	No

VPC-A is the primary network boundary for the project. Its 10.0.0.0/16 address space is divided into six /24 subnets across three Availability Zones.

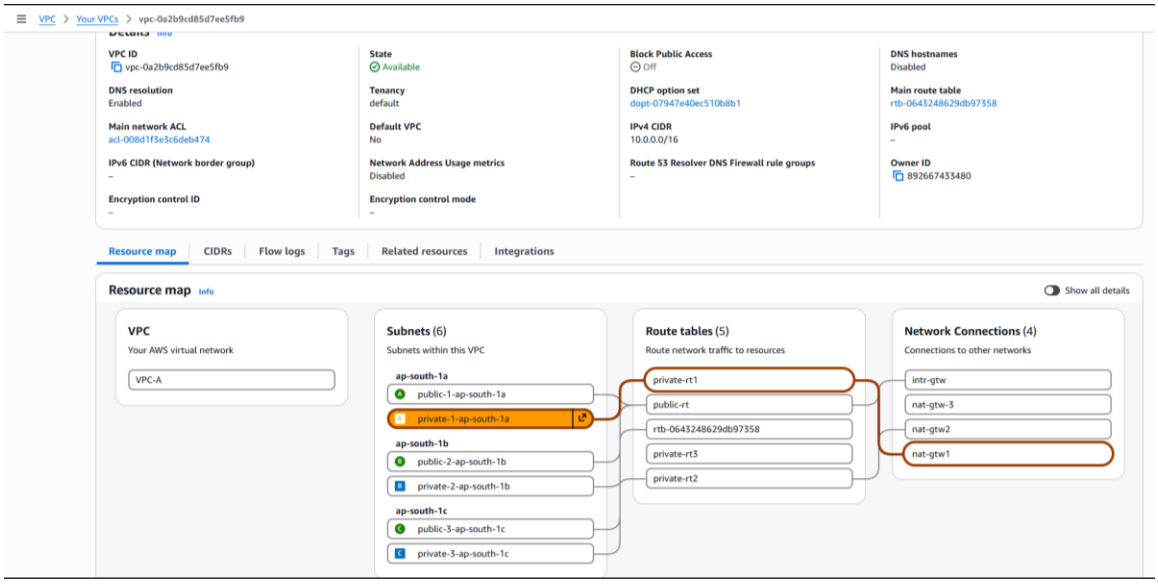
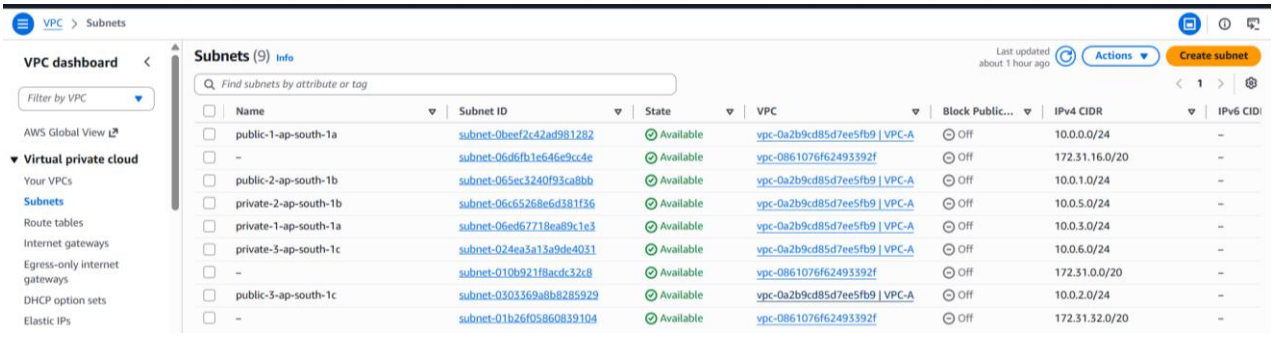


Figure : VPC-A overview and resource map.

5. Availability Zones & Subnets

AZ	Public subnet	Private subnet
ap-south-1a	public-1-ap-south-1a	private-1-ap-south-1a
ap-south-1b	public-2-ap-south-1b	private-2-ap-south-1b
ap-south-1c	public-3-ap-south-1c	private-3-ap-south-1c
Subnet	**CIDR**	**Role**
public-1-ap-south-1a	10.0.0.0/24	Public
public-2-ap-south-1b	10.0.1.0/24	Public
public-3-ap-south-1c	10.0.2.0/24	Public
private-1-ap-south-1a	10.0.3.0/24	Private
private-2-ap-south-1b	10.0.5.0/24	Private
private-3-ap-south-1c	10.0.6.0/24	Private



Six project subnets distributed across three Availability Zones.

6. Internet Gateway & Public Route

Component	Configuration
Internet Gateway	intr-gtw
IGW ID	igw-0e767b9e8d346e46c
State	Attached
Public route table	public-rt
Public subnet associations	3

The public route table contains the local VPC route and a default route (0.0.0.0/0) to the Internet Gateway. This makes the associated subnets part of the public networking layer.

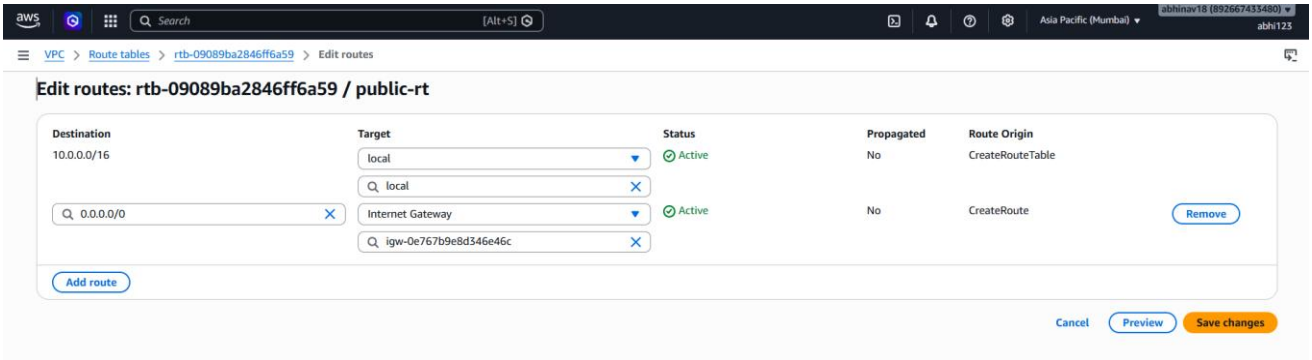
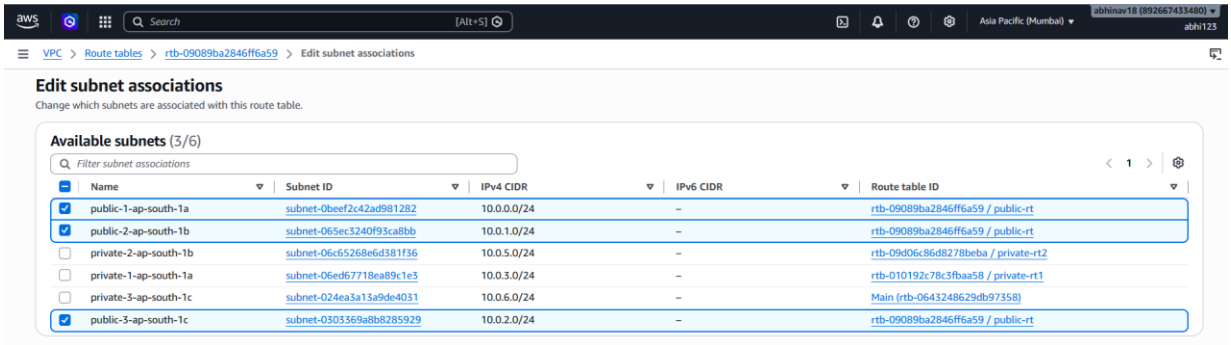


Figure : public-rt default route 0.0.0.0/0 points to intr-gtw.



Internet Gateway intr-gtw attached to VPC-A.

7. Route Table Configuration

Route table	Associated subnet(s)	Purpose
public-rt	3 public subnets	Public Internet routing
private-rt1	private-1-ap-south-1a	Private AZ-1 routing
private-rt2	private-2-ap-south-1b	Private AZ-2 routing
private-rt3	private-3-ap-south-1c	Private AZ-3 routing

The VPC also contains the AWS main/default route table. The custom route tables provide clear separation between public and private routing domains.

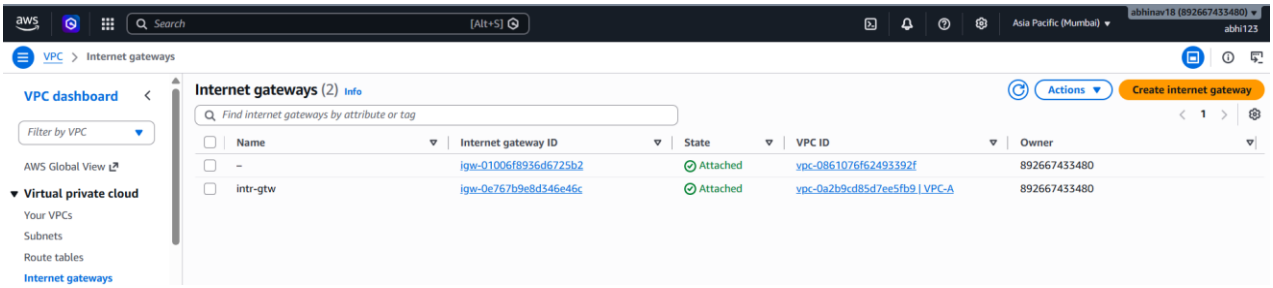
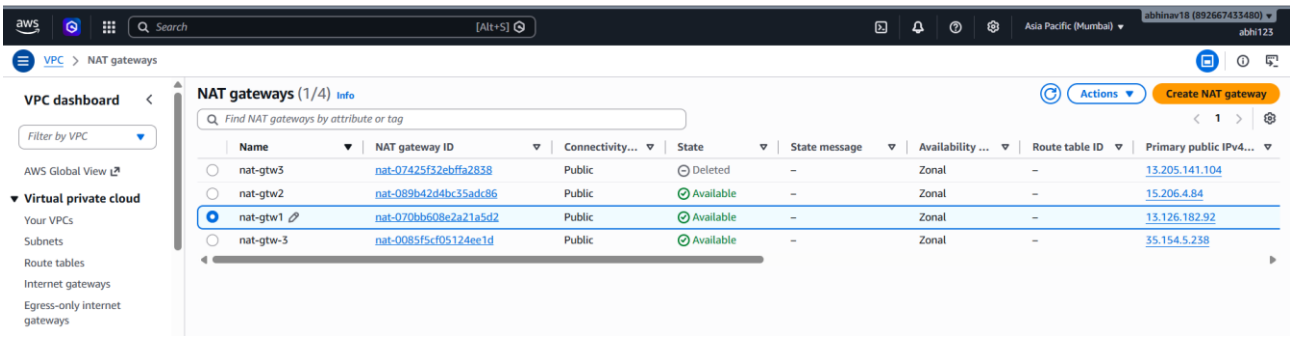


Figure 7. Route-table inventory for VPC-A.

8. NAT Gateway Configuration

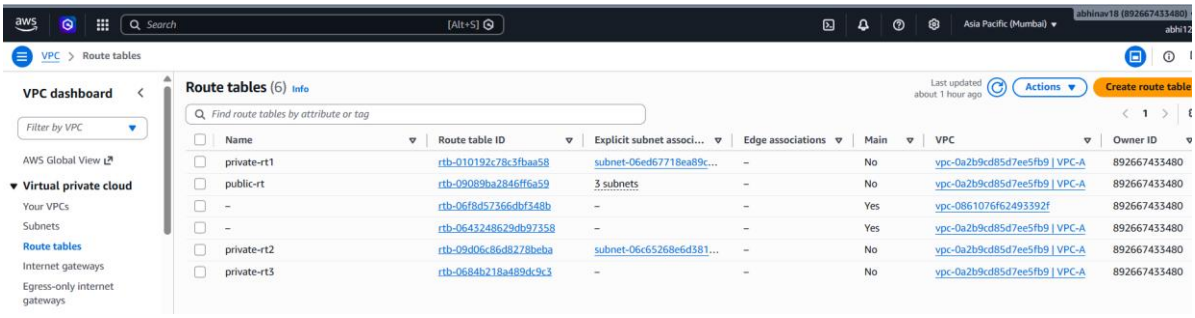
Private subnet	NAT Gateway	Availability
private-1-ap-south-1a	nat-gtw1	Available
private-2-ap-south-1b	nat-gtw2	Available
private-3-ap-south-1c	nat-gtw3	Available

Each private subnet has an AZ-specific NAT Gateway path. The supplied console evidence shows three project NAT Gateways in Available state.



NAT Gateway inventory showing the project NAT Gateways.

Elastic IP addresses are allocated for the NAT/public networking layer. The project console showed seven Elastic IP addresses in the Region; the screenshot alone does not establish which individual addresses belong to each NAT Gateway.

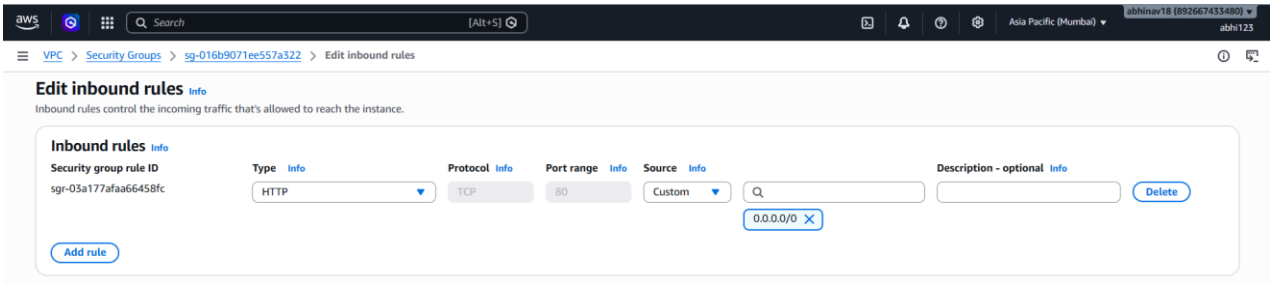


Elastic IP address inventory.

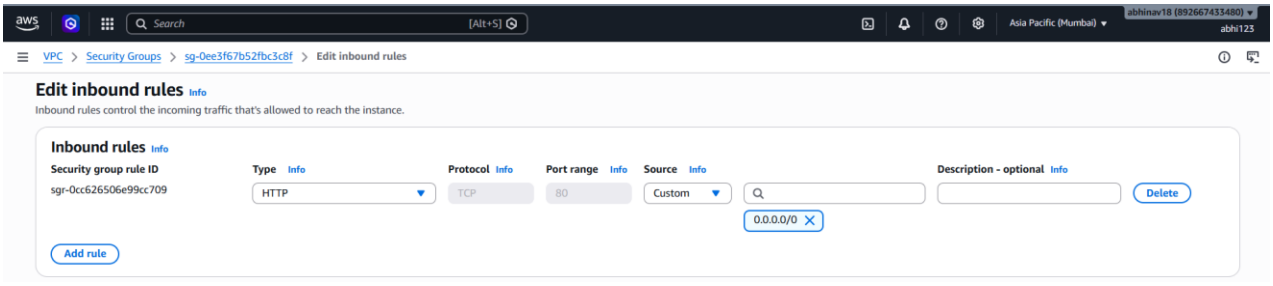
9. Security Groups

Security Group	ID	Observed inbound rule
ALB S.G.	sg-016b9071ee557a322	HTTP / TCP 80 from 0.0.0.0/0
EC2 S.G.	sg-0ee3f67b52fbc3c8f	HTTP / TCP 80 from 0.0.0.0/0

The screenshots supplied for the inbound rules confirm HTTP port 80 from 0.0.0.0/0 for both project security groups. Outbound rules are not shown in the supplied evidence and are therefore not documented here.



ALB S.G. inbound rule: HTTP port 80 from 0.0.0.0/0.

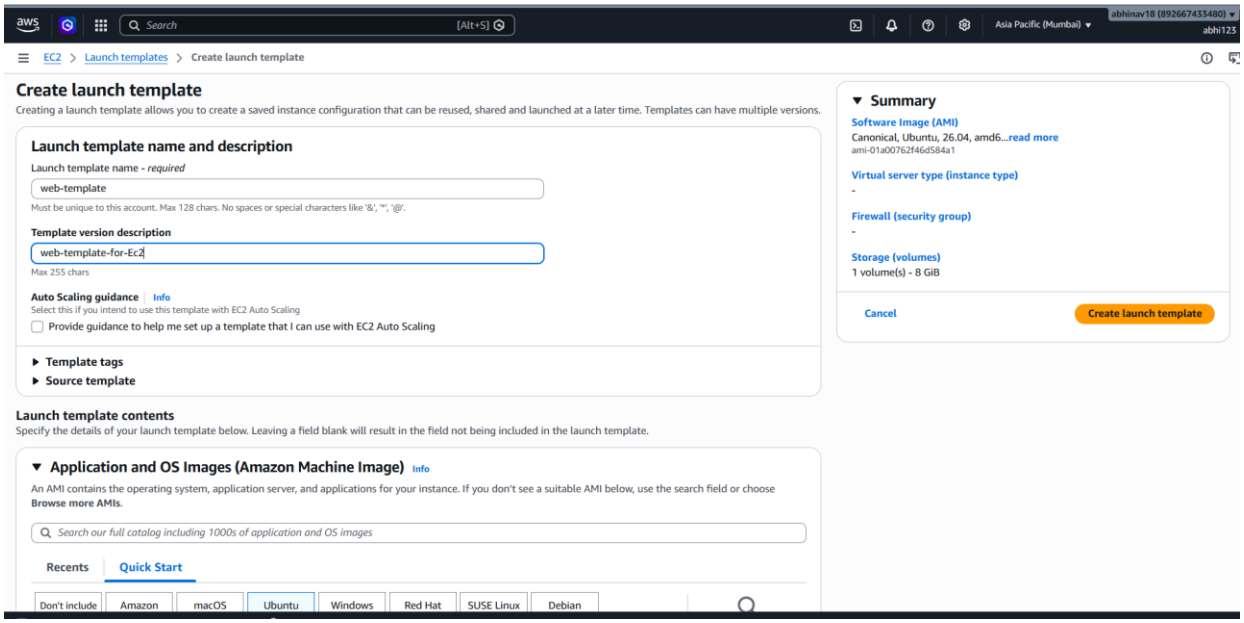


EC2 S.G. inbound rule: HTTP port 80 from 0.0.0.0/0.

10. EC2 Launch Template

Property	Actual value
Launch Template	web-template
Launch Template ID	lt-0e8870ecab5f03102
Default version	1
Latest version	1
AMI	ami-01a0076246d584a1
Operating system	Canonical Ubuntu 24.04
Instance type	t3.micro
Storage	8 GiB
Key pair	Not visible in supplied evidence
User Data	Not visible in supplied evidence

The Launch Template provides a reusable definition for instances launched by the Auto Scaling Group. The evidence confirms the Ubuntu 24.04 image, t3.micro type and 8 GiB volume.

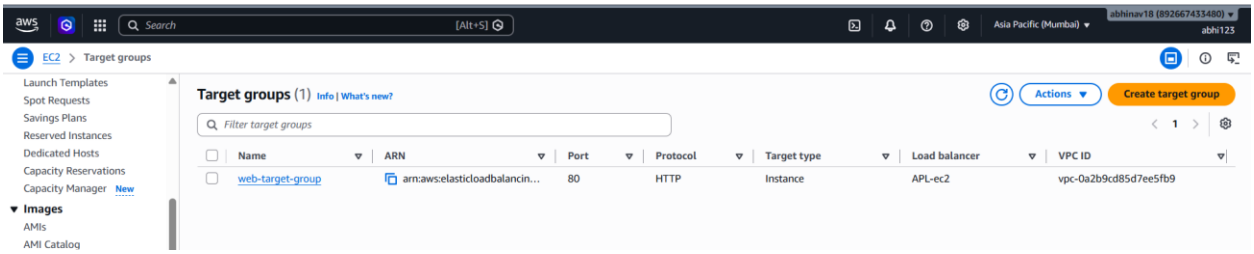


web-template Launch Template configuration.

11. Target Group

Property	Actual value
Name	web-target-group
Target type	Instance
Protocol	HTTP
Port	80
Load balancer	APL-ec2
VPC	VPC-A
Health check	[Add exact health-check settings if required]
Registered targets	[Add target list if required]
Target health	[Add current status if required]

The Target Group is the backend destination used by the ALB. It defines the target type and the protocol/port used for communication between the load balancer and the EC2 application servers.



web-target-group configured for instance targets on HTTP port 80.

12. Application Load Balancer

Property	Actual value
Name	APL-ec2
Type	Application
Status	Active
Scheme	Internet-facing
IP address type	IPv4
VPC	VPC-A
Availability Zones	ap-south-1a, ap-south-1b, ap-south-1c
Public subnets	3
DNS name	APL-ec2-67973852.ap-south-1.elb.amazonaws.com

APL-ec2 is the public-facing entry point of the application. It is active, Internet-facing, IPv4-based, and mapped to one public subnet in each of the three Availability Zones.

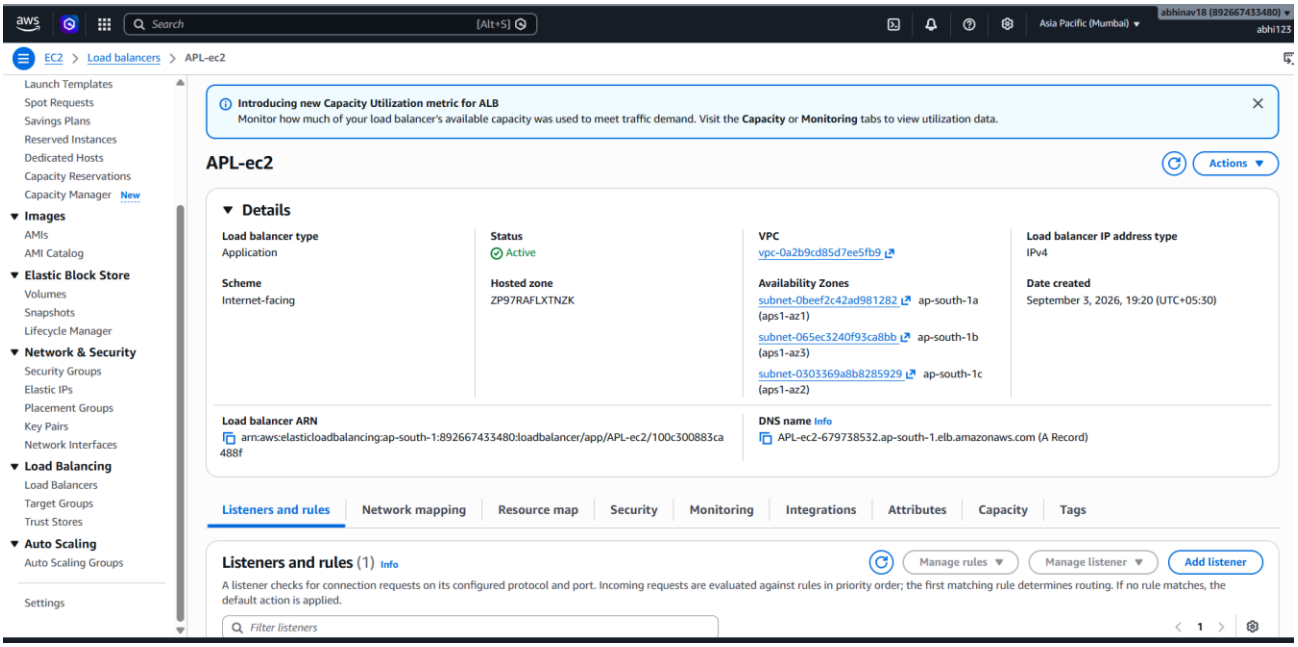


Figure : APL-ec2 ALB details, DNS name and three Availability Zones.

13. ALB Listener & Routing

The ALB console shows one configured listener. The supplied architecture and Target Group evidence identify HTTP port 80 as the application traffic path. The detailed listener row/rule is below the visible portion of the supplied ALB screenshot, so the exact listener rule action is not assumed beyond the documented Target Group association.

Client → APL-ec2 → Listener → web-target-group → EC2

Listener item	Evidence status
Listener count	1
Protocol/port	[Confirm from listener row]
Default action	[Confirm from listener row]
Forward target	web-target-group

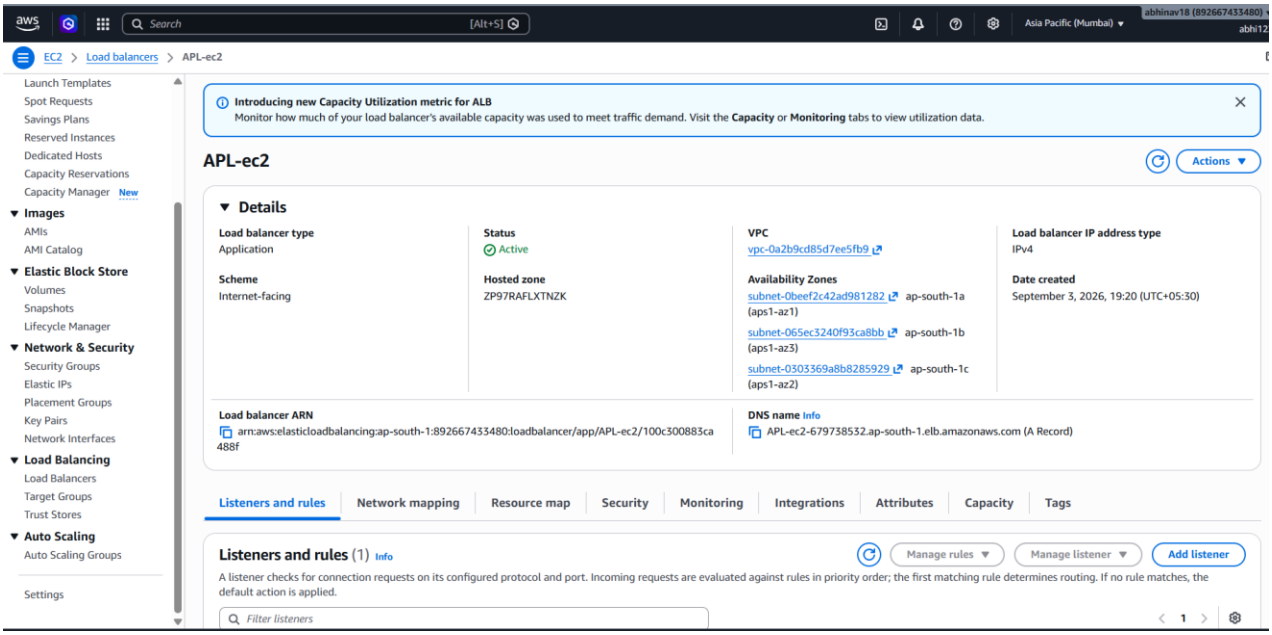


Figure : ALB console showing the Listeners and rules section.

14. Auto Scaling Group

Property	Observed value
ASG name	web-auto-scaling-group
Launch Template	web-template
Version	1
Instance type	t3.micro
Desired capacity at creation	1
Minimum capacity at creation	1
Maximum capacity at creation	1
Scaling policy at creation	No scaling policies

The ASG creation evidence shows a starting desired/minimum/maximum capacity of 1 and no scaling policy selected at that point. This is creation-time evidence and should not be interpreted as the current ASG configuration after later changes.

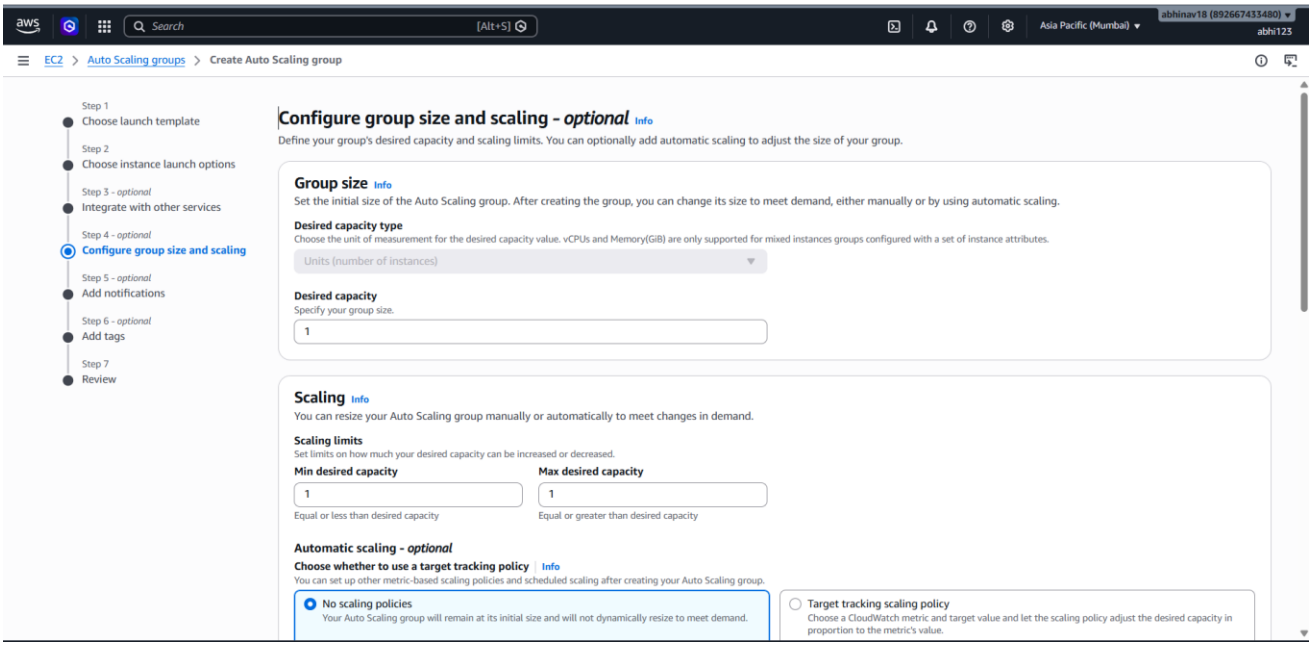


Figure : Auto Scaling Group creation-time capacity settings: desired 1, minimum 1, maximum 1.

15. EC2 Instances & Availability

The current EC2 console evidence shows four running t3.micro instances. They are distributed across ap-south-1a, ap-south-1b and ap-south-1c, and all displayed instances have passed 3/3 status checks.

The screenshot establishes the current EC2 console state. It does not, by itself, prove that every listed instance is currently registered in the Target Group or managed by the ASG.

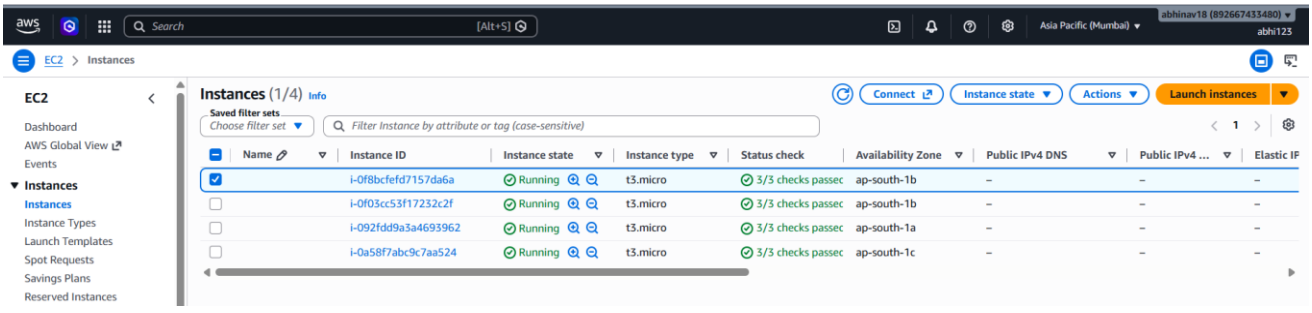


Figure : Current EC2 instance inventory: four running t3.micro instances across three Availability Zones.

16. Relationship Between ALB, Target Group, ASG & EC2

Component	Responsibility
ALB (APL-ec2)	Receives client traffic and selects backend targets.
Listener	Accepts connections and applies routing rules.
Target Group	Maintains the backend target set and health-check state.
ASG	Maintains the desired EC2 fleet according to its configured capacity/policies.
Launch Template	Defines the reusable EC2 launch configuration.
EC2 instances	Run the web application and receive forwarded requests.

APL-ec2 → Listener → web-target-group → Healthy EC2

The Target Group is the integration point: the ALB forwards requests to it, while the Auto Scaling Group can register/deregister instances as the managed fleet changes.

17. Complete Request Flow

USER
↓
Internet
↓
Internet Gateway
↓
Application Load Balancer (APL-ec2)
↓
Listener
↓
web-target-group
↓
Healthy EC2 instance
↓
Application response

Why this flow works

- The public route sends Internet-bound traffic through the Internet Gateway.
- The ALB is placed in public subnets across three Availability Zones.
- The listener determines the forwarding action.
- The Target Group provides the backend targets.
- Only eligible healthy targets should receive normal forwarded traffic.

Backend response

The selected EC2 instance processes the HTTP request and returns the response through the established connection path back to the client.

18. Private EC2 Outbound Internet Connectivity

Private EC2
↓
Private Route Table
↓
NAT Gateway
↓
Internet Gateway
↓
Internet

AZ	Private route table	NAT Gateway
ap-south-1a	private-rt1	nat-gtw1
ap-south-1b	private-rt2	nat-gtw2
ap-south-1c	private-rt3	nat-gtw3

The private instances do not need public IP addresses for outbound Internet access. The NAT Gateway translates the private source address for Internet-bound connections and uses the public Internet path through the Internet Gateway.

19. High Availability, Health Checks & Scaling

High availability

- Three Availability Zones reduce dependence on a single AZ.
- The ALB is mapped to public subnets in all three AZs.
- Private subnets are distributed across all three AZs.
- Each private subnet has a dedicated NAT Gateway and route table.

Target health checks

The Target Group periodically checks configured targets using its health-check settings. An unhealthy target is not considered an eligible normal destination for new load-balanced traffic while it remains unhealthy.

Auto Scaling

The Auto Scaling Group uses web-template to create consistent EC2 instances. The supplied creation screenshot shows no scaling policy at that time, so dynamic scale-out/scale-in behavior should only be claimed if later evidence is captured.

20. Testing & Validation

Test	Evidence to capture	Current documentation status
ALB DNS / application	Browser showing ALB DNS response	[Add actual result]
Load balancing	Responses demonstrating backend distribution	[Add actual result]
Target health	Target Group target-health screen	[Add actual result]
Auto Scaling	ASG activity / desired vs running capacity	Creation settings evidenced; result not yet evidenced
Failure test	One backend made unhealthy	[Add actual result]
Continuity	Application response after unhealthy target	[Add actual result]
NAT connectivity	Private EC2 outbound Internet test	[Add actual result]

Important: this report does not mark unperformed tests as successful. The final testing evidence should be inserted only after the corresponding activity is actually performed.

Recommended test evidence

- ALB DNS opening the application.
- Target Group showing healthy targets.
- A controlled unhealthy-target test followed by an application request.
- Private EC2 command such as curl to a public endpoint, if performed.
- ASG activity/history showing any capacity change, if a scaling event is tested.

21. Project Result, Benefits, Cost & Conclusion

Complete architecture summary

VPC-A (10.0.0.0/16) in ap-south-1 contains six subnets across three Availability Zones. Three public subnets support the Internet-facing ALB. Three private subnets support the application layer and have dedicated NAT Gateway paths. A Launch Template, Target Group, ALB and Auto Scaling Group provide the application delivery and instance-management components.

Benefits

- Multi-AZ architecture
- Centralized application entry point
- Health-aware backend routing
- Private application subnet design
- Reusable EC2 configuration
- AZ-specific NAT connectivity

Cost considerations

The main cost components in this lab architecture can include the Application Load Balancer, EC2 usage, three NAT Gateways, public IPv4 addresses and data transfer. For a learning environment, resources should be stopped or deleted when no longer required and billing should be monitored.

What I learned

- VPC and public/private subnet design.
- Route tables and Internet Gateway routing.
- NAT Gateway operation for private workloads.
- Security Group traffic control.
- ALB, listener and Target Group integration.
- Launch Template and Auto Scaling concepts.
- Multi-AZ availability planning.

Final project result

The hands-on project successfully establishes the documented AWS infrastructure for a multi-AZ web application architecture. The supplied evidence confirms the core network, load-balancing and EC2 components. Application and failure-testing

outcomes remain evidence-driven and should be added when those tests are captured.

Conclusion

This project demonstrates practical AWS networking and application delivery skills by combining VPC design, public/private routing, NAT Gateways, an Internet-facing ALB, Target Group health management, EC2 Launch Templates and Auto Scaling. The architecture is suitable as a fresher Cloud/DevOps portfolio project because it demonstrates how individual AWS services work together as one deployable system.